Duc-Cuong Vu

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Researcher in nonlinear control and robotics, with a focus on safety-critical control, model predictive control, and dynamic robotic systems.

Education

Master of Sci. in Automation and Control, *Hanoi University of Science and Technology (HUST)*, *Jul 2024 - present*
Funded by: *Master, PhD Scholarship Programme of Vingroup Innovation Foundation (VinIF)*

Bachelor of Sci. in Automation and Control, *Hanoi University of Science and Technology (HUST)*, *Oct 2020 - Mar 2024*
Excellent degree, GPA: 3.71/4 (Rank: 22/499, top 5%). Finished the 4-year BSc program in just 3.5 years.

Selected publications

- ⋆ Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks
  Ocean Engineering (SCIE Q1) (2025) [pdf]
  D.C. Vu†, *S. Tran*†, *T.L. Nguyen, and D.C. Hoang* †: *equally contribution*
- ⋆ Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations
  Ocean Engineering (SCIE Q1) (2025) [pdf]
  D.C. Vu, *D.H. Nguyen, M.N. Vu, and T.L. Nguyen*
- ⋆ CBFs-based Model Predictive Control for Obstacle Avoidance with Tilt Angle Limitation for Ball-Balancing Robots
  IEEE Acess (SCIE Q2) (2025) [pdf]
  M.D. Pham, ***D.C. Vu****, T.T.H. Nguyen, T.V.A. Nguyen, M.N. Vu, and T.L. Nguyen*
- ⋆ A novel approach of Consensus-based Finite-time DSMC for Stewart platform manipulators motion tracking
  Results in Engineering (ESCI Q1) (2025) [pdf]
  D.C. Vu*, D.H. Nguyen, and T.L. Nguyen*
- ⋆ Unifying Hierarchical Sliding Mode Control and Control Barrier Function for Tilt Angle Constraint of a Ball-Balancing Robot
  IET Cyber-Systems and Robotics (SCIE Q3) (2025) [pdf]
  T.T.H. Nguyen, ***D.C. Vu****, M.D. Pham, T.V.A. Nguyen, and T.L. Nguyen*
- ⋆ Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints
  International Journal of Robust and Nonlinear Control - IJRNC (SCIE Q1) (2024) [pdf]
  D.C. Vu*, M.D. Pham, T.T.H. Nguyen, T.V.A. Nguyen, and T.L. Nguyen*

Other manuscripts

- ⋆ Decentralized Control Strategy with Consistent Joint-to-Actuator Control Law for Parallel Mechanisms on a Humanoid Platform
  Submitted manuscript [pdf]
  T.V. Trung, V.T. Nguyen, ***D.C. Vu****, D.H. Nguyen, M.C. Nguyen, and M.N. Vu*
- ⋆ Adaptive Whole-Body MPC with Optimal Posture Planning for Humanoid Robots under Unknown Payloads
  Submitted manuscript
  M.C. Nguyen†, ***D.C. Vu***†, *V.A. Vu, T. Nguyen, D.T. Pham, M.H. Phung, A.V. Ngo, and M.N. Vu* †: *equally contribution*
- ⋆ Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators
  Submitted manuscript
  D.C. Vu*, V.T. Nguyen, D.H. Nguyen, M.C. Nguyen, T.V. Trung, A. Kugi, and M.N. Vu*

Experience

Robotics Engineer at VinRobotics *Sep 2025 - present*
Full-time, supervised by PhD. Minh Nhat Vu *Hanoi, Vietnam*

- ⋆ Developed standardized pipelines for robot modeling, kinematic calibration, and system identification for industrial robotic systems.
- ⋆ Designed and implemented motion planning and control algorithms for humanoid robots.
- ⋆ Contributed to research activities, with papers submitted to *IROS 2026* and *IEEE-TASE*, *IEEE RA-L*.

Research Project Assistant at HUST *Jan 2025 - present*
Contract (Hybrid), supervised by PhD. Chinh Hoang Duc and Assoc.Prof.PhD. Tung Lam Nguyen *Hanoi, Vietnam*

- ⋆ Contributed to a project on autonomous robot navigation integrating sensor networks and wireless communication.
- ⋆ Developed a MuJoCo-based simulation framework for AUVs, incorporating underwater dynamics, sensing, disturbances, and communication constraints.
- ⋆ Implemented and validated control algorithms for navigation, obstacle avoidance, and trajectory tracking.
- ⋆ Collaborated on system integration and co-authored a peer-reviewed scientific publication.

Research Project Assistant at HUST *Jan 2025 - Oct 2025*

Contract (Hybrid), supervised by PhD. Minh Nhat Vu and Assoc.Prof.PhD. Tung Lam Nguyen *Hanoi, Vietnam*

- ⋆ Worked on advanced control of a ship-mounted Stewart platform for marine applications (funded by KIST).
- ⋆ Designed robust and safety-critical control strategies for parallel robotic systems under disturbances.
- ⋆ Conducted modeling, simulation, and validation for nonlinear dynamic systems.

Graduate Researcher at HUST *May 2024 - present*

Part-time (Hybrid), supervised by Assoc.Prof.PhD. Tung Lam Nguyen *Hanoi, Vietnam*

- ⋆ Conducted research on control of parallel robotic systems for maritime applications.
- ⋆ Designed and implemented advanced control algorithms for a Stewart platform, validated through high-fidelity simulations (Simscape, MuJoCo).
- ⋆ Built experimental setups including mechanical assembly, hardware integration, Linux real-time systems, and EtherCAT communication.
- ⋆ Collaborated on publications and presented results to academic and industrial partners.

Student Intern at HUST *Oct 2021 - Apr 2024*

Internship (Hybrid), supervised by Assoc.Prof.PhD. Tung Lam Nguyen *Hanoi, Vietnam*

- ⋆ Developed mathematical models and simulation environments for 3D ballbot systems, focusing on nonlinear dynamics and trajectory generation.
- ⋆ Investigated advanced control strategies and implemented them in simulation and experimental setups.
- ⋆ Authored and co-authored peer-reviewed journal papers, including publications in International Journal of Robust and Nonlinear Control and IEEE Access.

Academic activities

Invited review for

- ⋆ *Nonlinear Dynamics*
- ⋆ *International Conference on Intelligent Robots and Systems (IROS, 2026)*
- ⋆ *IEEE Robotics and Automation Letters (RA-L)*

Teaching, Seminars and Talks

2025: Talk "MuJoCo for Advanced Physics Simulation: From manipulators to autonomous vehicles" for "Motion Control" master course at HUST and MoCAR seminar [pdf]

2025: Seminar "Underwater Vehicles" for modeling training of Autonomous Underwarter Vehicle at MEG-MoCAR [pdf]

Honours & awards

Master, PhD Scholarship Programme *Vingroup Innovation Foundation (VINIF)*

Best Thesis Defense Award *Hanoi University of Science and Technology*

Skills

Programming	Python, C/C++, MATLAB
Robotics & Control	Nonlinear control, Model Predictive Control (MPC), Control Barrier Functions (CBF), motion planning, system modeling and identification
Tools & Frameworks	ROS2, MuJoCo, Simulink/Simscape, Pinocchio, CasADi, Acados, VAMP, CAPT, Git
Systems & Implementation	Real-time Linux, IgH EtherCAT, hardware integration, motion capture systems
Research	Scientific writing, peer-reviewed publications, technical presentations, literature review, experimental validation